

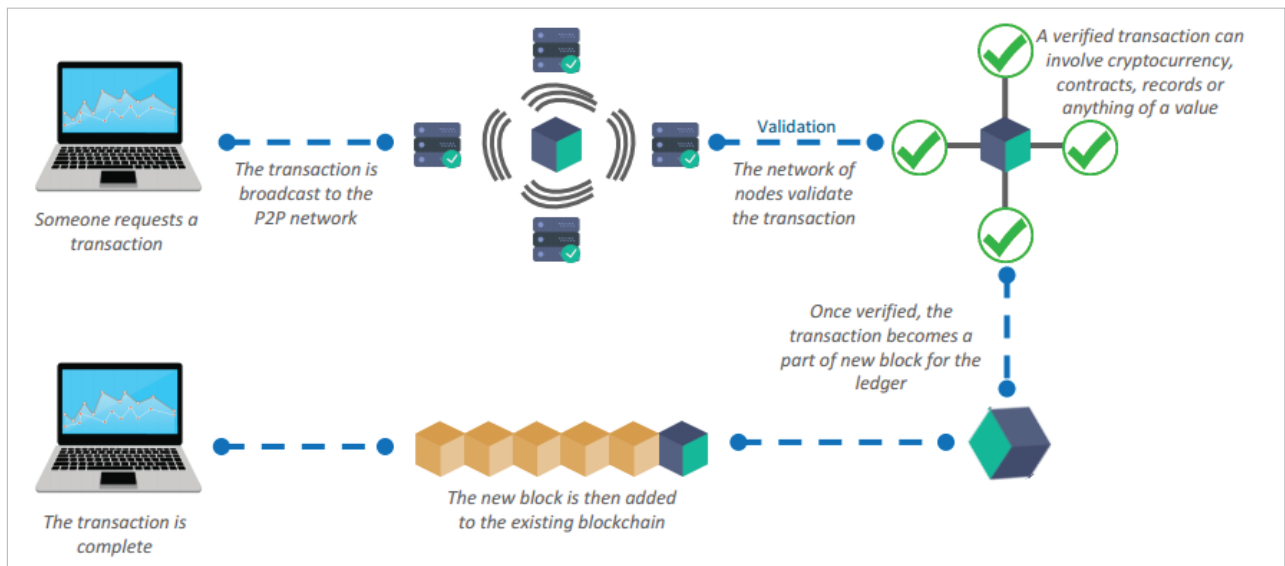


Figure 2: How the blockchain works

community of individuals collectively act as governing authorities. These individuals employ digital signatures to achieve consensus on various facets.

Upon receiving authorisation, a transaction undergoes meticulous mathematical verification, culminating in a successful and secure transaction between the interconnected parties within the network. In essence, blockchain users wield cryptography keys as they engage in a myriad of digital interactions across the peer-to-peer network.

Here's a use case that illustrates how blockchain works.

**Hash encryption:** The SHA256 algorithm secures data through hashing and encryption in blockchain technology. It transmits information about the sender's address (public key), the recipient's address, the transaction, and his or her private key. After verification, the hashed data is sent across the globe and added to the blockchain in encrypted form. The sender and receiver's authentication is made easier by the SHA256 algorithm's virtually impenetrable hash encryption.

**Proof of work:** Each block in a blockchain has four main headers.

- **Previous hash:** The previous block can be found at this hash address.
- **Transaction details:** Information about each transaction that must take place.
- **Nonce:** In cryptography, an arbitrary number is used to distinguish the block's hash address.
- **Hash address of the block:** A hashing algorithm transmits the preceding hash, transaction information, and nonce. The output value with a length of 64 characters and 256 bits is referred to as the distinct 'hash address' or block hash.

Numerous people around the world try to figure out the right hash value to meet a predetermined condition using computational algorithms. The transaction completes when the predetermined condition is met. In simpler terms, blockchain miners solve a mathematical puzzle, known as a proof of work problem. Whoever solves it first gets a reward.

**Mining:** The act of appending transactional data to the existing digital public ledger is commonly known as 'mining'. Although this term is closely linked to Bitcoin, it extends its application to other

blockchain technologies. The mining process entails the creation of a block transaction's hash, a highly secure and tamper-resistant mechanism. This process safeguards the entire blockchain network, eliminating the requirement for a centralised system.

## Types of blockchains

There are primarily three types of blockchains.

**Public blockchain:** Public blockchains are open and permissionless, meaning anyone can participate in the network, validate transactions, and create new blocks. They are decentralised and rely on a consensus mechanism, usually proof of work (PoW) or proof of stake (PoS), to validate and add new blocks. Bitcoin and Ethereum are examples of public blockchains. Public blockchains provide transparency and security but may have scalability limitations due to their openness.

**Private blockchain:** Private blockchains, also known as permissioned blockchains, are restricted to a specific group of participants. These participants are known and trusted, and they are given permission to access and